

# General Law of Conservation of Energy

SS ILP Physics (Intermediate), Week 4 Class 2



# Outline

- Conservation of Mechanical Energy (Recap, with conservative force)
- General Law of Conservation of Energy (with non-conservative force)

# Conservation of Mechanical Energy (Recap)

- Recall from previous lesson, the total mechanical energy is the summation of kinetic energy and potential energy:

$$E = K + U$$

- If the net work done is only due to conservative force(s), total mechanical energy of an isolated system of objects is constant. It means that kinetic and potential energy simply converts into one another in the system.

$$E_i = E_f \quad \longleftrightarrow \quad K_i + U_i = K_f + U_f$$

- Or, if more than one type of potential energy involved

$$K_i + \sum U_i = K_f + \sum U_f$$

# Conservation of Mechanical Energy (Gravitational forces only, Recap)

From work-energy theorem,

$$W_{total} = \Delta K$$

We obtain

$$\Delta K = W_{total} = W_{grav} = -\Delta U_{grav}$$

$$K_2 - K_1 = -(U_{grav,2} - U_{grav,1})$$

$$K_1 + U_{grav,1} = K_2 + U_{grav,2}$$

$$\frac{1}{2}mv_1^2 + mgy_1 = \frac{1}{2}mv_2^2 + mgy_2$$

Total mechanical energy  
at position  $y_1$



Total mechanical energy  
at position  $y_2$

## Conservation of mechanical energy:

When only the gravity does work, the total mechanical energy of the system is conserved:

$$E = K + U_{grav} = \text{constant}$$

H.D. Young and R. A. Freedman, Sears and Zemansky's  
University Physics. 13E, Addison-Wesley, Boston, 2012.

# Conservation of Mechanical Energy (Spring forces only, Recap)

From work-energy theorem,

$$W_{total} = \Delta K$$

We obtain

$$\Delta K = W_{total} = W_{el} = -\Delta U_{el}$$

$$K_2 - K_1 = -(U_{el,2} - U_{el,1})$$

$$K_1 + U_{el,1} = K_2 + U_{el,2}$$

$$\frac{1}{2}mv_1^2 + \frac{1}{2}kx_1^2 = \frac{1}{2}mv_2^2 + \frac{1}{2}kx_2^2$$

Total mechanical energy  
at position  $x_1$



Total mechanical energy  
at position  $x_2$

## Conservation of mechanical energy:

When only the spring force does work, the total mechanical energy of the system is conserved:

$$E = K + U_{el} = \text{constant}$$

H.D. Young and R. A. Freedman, Sears and Zemansky's  
University Physics. 13E, Addison-Wesley, Boston, 2012.

# Conservative Vs Non-conservative Force

- A conservative force allows conversion between kinetic and potential energy. Gravitational and the spring force are conservative.
- The work done between two points by any conservative force
  - I. is independent of the path between the two points.
  - II. is zero if the starting and ending points are the same.
  - III. is reversible.
  - IV. can be expressed in terms of a **potential energy function**. Note: only conservative force allows to define potential energy meaningfully due to property (I).
- A force (such as friction, air drag) that is not conservative is called a **non-conservative force**, or a *dissipative force*. Note: not all non-conservative forces are dissipative! e.g magnetic (Lorentz) force.

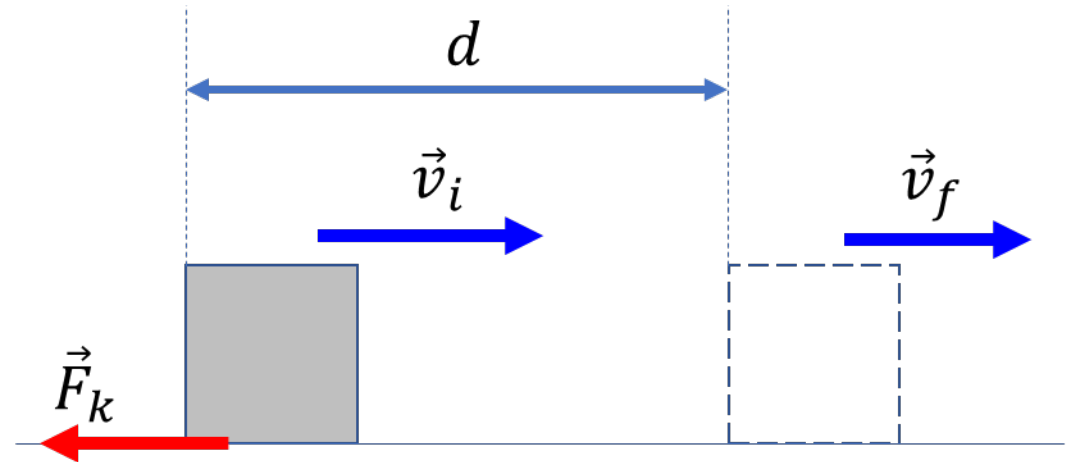
R.A. Serway and J.W. Jewett, Physics for Scientists and Engineers with Modern Physics. 9E, Cengage Learning, Boston, 2014.

# Non-conservative force: Kinetic Friction

- Suppose an object is given an initial velocity  $\vec{v}_i$  slides on a rough horizontal surface (experiences frictional force  $\vec{F}_k$ ) for a distance  $d$  before reaching a final velocity  $\vec{v}_f$ :

$$\begin{aligned}\Delta K &= K_f - K_i \\ &= \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = -|\vec{F}_k|d\end{aligned}$$

Can't be written as change of potential energy since the force is non-conservative



- Loss in KE is  $|\vec{F}_k|d$ , which is the energy dissipated by the force of kinetic friction. Part of the energy is transferred to the internal energy of the object, and part to the surface.

# Conservation of Mechanical Energy: General Form

- Work-energy theorem still works:

$$W_{total} = \Delta K$$

$$W_{grav} + W_{el} + W_{other} = \Delta K$$

$$-(U_{grav,2} - U_{grav,1}) - (U_{el,1} - U_{el,2}) + W_{other} = K_2 - K_1$$

- Rearrange:

$$K_1 + U_{grav,1} + U_{el,1} + W_{other} = K_2 + U_{grav,2} + U_{el,2}$$

- Or more compactly,

$$E_1 + W_{other} = E_2 \longrightarrow \boxed{W_{other} = E_2 - E_1}$$

where  $E_i = K_i + U_{grav,i} + U_{el,i}$  is the total mechanical energy of initial ( $i = 1$ ) and final ( $i = 2$ ) states.

The work done by all forces other than the gravitational force or elastic force equals the change in the total mechanical energy,  $E = K + U$ , of the system, where  $U = U_{grav} + U_{el}$  is the sum of the gravitational and elastic potential energy



# General Law of Conservation of Energy

- Friction and other non-conservative forces remove/add mechanical energy from/to the system. This is the amount by which the mechanical energy of the system changes, denoted by  $W_{\text{other}}$ . Thus,

$$K_i + U_i + W_{\text{other}} = K_f + U_f \quad \longleftrightarrow \quad E_i + W_{\text{other}} = E_f$$

- Example: In the case of kinetic friction and air resistance, since the force and displacement are always in opposite direction, the work done is always negative, and the final mechanical energy is always less than the initial mechanical energy (where  $d$  is distance travelled):

$$K_i + U_i - |\vec{F}_k|d = K_f + U_f \quad \longleftrightarrow \quad E_i - |\vec{F}_k|d = E_f$$

# General Law of Conservation of Energy

- Non-conservative force cannot represent by potential energy, but we can describe the effects of these forces in terms of kinds of energy other than kinetic or potential energy.
- Example, when a car with locked brakes skids to a stop, both the tires and the road surface become hotter. The energy associated with this change in the state of the materials is called **Internal Energy**  $U_{\text{int}}$ .
- Experimentally, we observed that change in internal energy is:

$$\Delta U_{\text{int}} = -W_{\text{other}}$$

H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

# General Law of Conservation of Energy

- We can write the law of conservation of energy as:

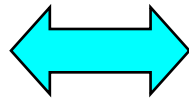
$$K_i + U_i - \Delta U_{\text{int}} = K_f + U_f \quad \longleftrightarrow \quad \Delta E + \Delta U_{\text{int}} = 0$$

Energy can never be created or destroyed. Energy may be transformed from one form to another, but the total energy of an isolated system is always constant!!!

# Strategy for applying energy conservation

- Choose initial & final points.
- Select a zero-reference point for each form of P.E., and write the corresponding expression of P.E.
- If the net work done is only due to conservative force(s), apply

$$E_i = E_f$$



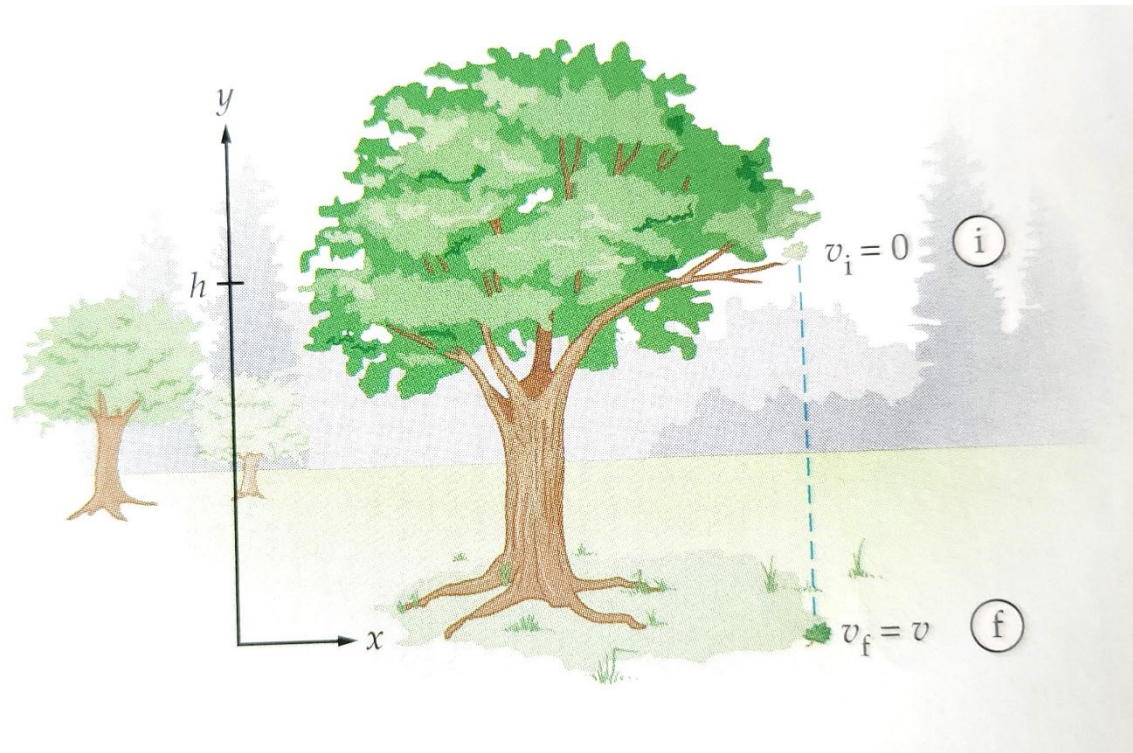
$$K_i + \sum U_i = K_f + \sum U_f$$

- If the net work done is (partly) due to non-conservative force(s) (like friction, air resistance), apply

$$E_i + W_{\text{other}} = E_f \quad \text{or} \quad \Delta E + \Delta U_{\text{int}} = 0$$

# Example

Deep in the forest, a 17.0-g leaf falls from a tree and drops straight to the ground. If its initial height was 5.30 m and its speed on landing was 1.3 m/s, how much nonconservative work was done on the leaf? Take  $g = 9.81 \text{ m/s}^2$ .



J. S. Walker, Physics, Example 8-9, p223

## Example: Solution

Deep in the forest, a 17.0-g leaf falls from a tree and drops straight to the ground. If its initial height was 5.30 m and its speed on landing was 1.3 m/s, how much nonconservative work was done on the leaf? Take  $g = 9.81 \text{ m/s}^2$ .

Apply the general law of conservation of energy,  $E_i + W_{\text{other}} = E_f$

$$E_i = K_i + U_i = 0 + mgh = mgh$$

$$E_f = K_f + U_f = \frac{1}{2}mv^2 + 0 = \frac{1}{2}mv^2$$

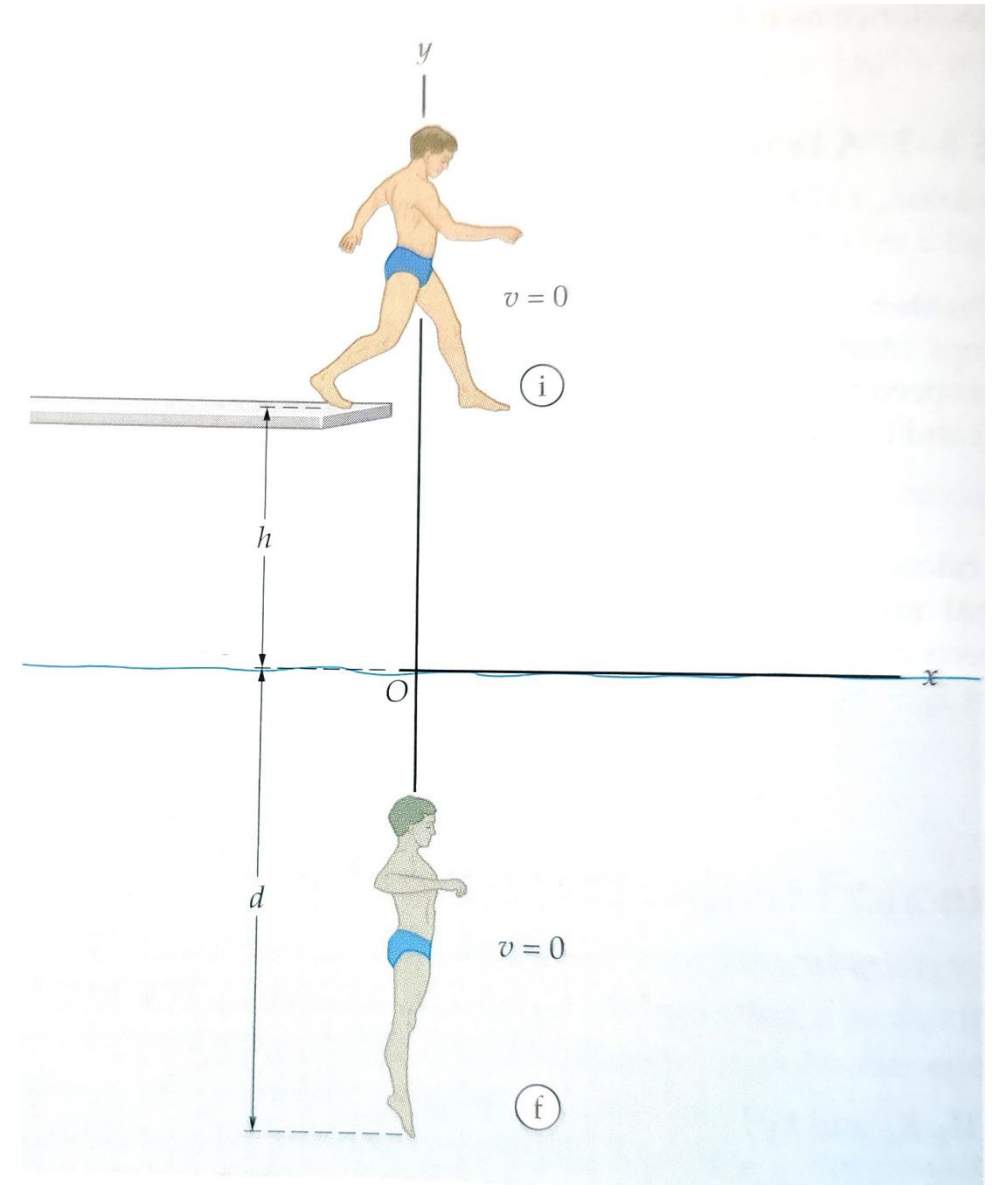
$$W_{\text{other}} = W_{nc}$$

$$W_{nc} + mgh = \frac{1}{2}mv^2$$

$$W_{nc} = -0.87 \text{ J}$$

# Case Problem 1

A 95.0-kg diver steps off a diving board and drops into the water 3.00 m below. At some depth  $d$  below the water's surface, the diver comes to rest. If the nonconservative work done on the diver is  $W_{nc} = -5120 \text{ J}$ , what is the depth,  $d$ ? Take  $g = 9.81 \text{ m/s}^2$ .



J. S. Walker, Physics, Active Example 8-2, p224

# Case Problem 1: Solution

A 95.0-kg diver steps off a diving board and drops into the water 3.00 m below. At some depth  $d$  below the water's surface, the diver comes to rest. If the nonconservative work done on the diver is  $W_{nc} = -5120 \text{ J}$ , what is the depth,  $d$ ? Take  $g = 9.81 \text{ m/s}^2$ .

Apply the general law of conservation of energy,  $E_i + W_{\text{other}} = E_f$

$$E_i = K_i + U_i = 0 + mgh = mgh$$

$$W_{nc} + mgh = -mgd$$

$$E_f = K_f + U_f = mg(-d) + 0 = -mgd$$

$$d = 2.49 \text{ m}$$

$$W_{\text{other}} = W_{nc}$$



## Case Problem 2:

An 80.0-kg jogger starts from rest and runs uphill into a stiff breeze. At the top of the hill the jogger has done the work  $W_{nc1} = 1.80 \times 10^4 J$ , air resistance has done the work  $W_{nc2} = -4420 J$ , and the jogger's speed is 3.50 m/s. Find the height of the hill.



J. S. Walker, Physics, Active Example 8-3, p227

## Case Problem 2: Solution

An 80.0-kg jogger starts from rest and runs uphill into a stiff breeze. At the top of the hill the jogger has done the work  $W_{nc1} = 1.80 \times 10^4 J$ , air resistance has done the work  $W_{nc2} = -4420 J$ , and the jogger's speed is 3.50 m/s. Find the height of the hill.

Apply the general law of conservation of energy,  $E_i + W_{\text{other}} = E_f$

$$E_i = K_i + U_i = 0 + 0 = 0$$

$$E_f = K_f + U_f = mgh + \frac{1}{2}mv^2$$

$$W_{\text{other}} = W_{nc1} + W_{nc2}$$

$$W_{nc1} + W_{nc2} = mgh + \frac{1}{2}mv^2$$

$$h = 16.7 \text{ m}$$

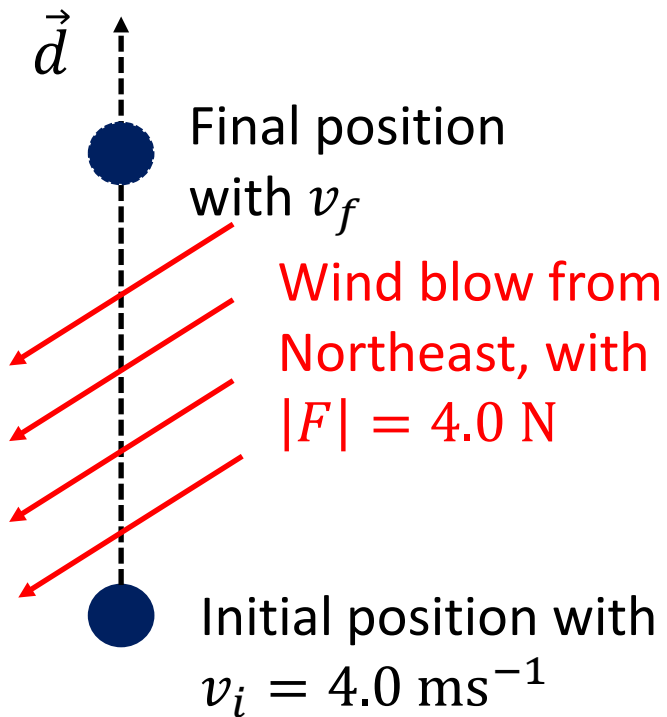
## Example

A 50 kg ice skater is gliding along the ice, heading due north at  $4.0 \text{ ms}^{-1}$ . The ice has a small coefficient of static friction, to prevent the skater from slipping sideways, but with zero kinetic friction along her path. Suddenly, a wind from the northeast exerts a force of 4.0 N on the skater. Use conservation of energy to find the skater's speed after gliding 100 m in this wind.

R.D. Knight, Physics for Scientists and Engineers: A Strategic Approach. 4E, Pearson, Boston, 2016.

# Example: Solution

There is no kinetic friction along her direction of motion. Static friction acts to prevent her skates from slipping sideways on the ice, but this force is perpendicular to the motion and does not contribute to the change of mechanical energy. There is no change of gravitational P.E. since the motion doesn't involve difference in height.



$$W_{\text{other}} = \vec{F} \cdot \vec{d} = |\vec{F}| |\vec{d}| \cos 135^\circ = -282.8 \text{ J}$$

$$\text{From COE, } K_i + U_i + W_{\text{other}} = K_f + U_f$$

$$K_i - K_f = -W_{\text{other}} \Rightarrow \frac{1}{2} m (v_i^2 - v_f^2) = 282.8 \text{ J}$$

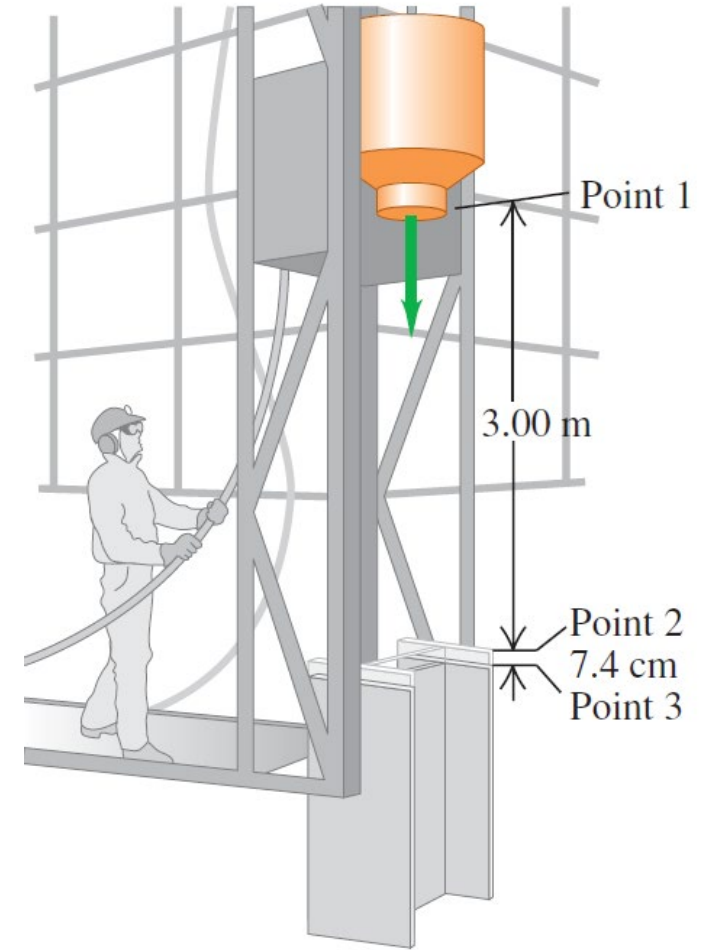
$$\Rightarrow v_f^2 = v_i^2 - 2(282.8 \text{ J})/m$$

$$\therefore v_f = \sqrt{v_i^2 - 2 \frac{282.8 \text{ J}}{m}} = \sqrt{(4)^2 - \frac{2}{(50)} (282.8)} = 2.2 \text{ m} \cdot \text{s}^{-1}$$

## Case Problem 3

The 200 kg steel hammerhead of a pile driver is lifted 3.00 m above the top of a vertical I-beam being driven into the ground (see Figure). The hammerhead is then dropped, driving the I-beam 7.4 cm deeper into the ground. The vertical guide rails exert a constant 60-N friction force on the hammerhead. Find:

- (a) the speed of the hammerhead just as it hits the I-beam; and
- (b) the average force the hammerhead exerts on the I-beam.



H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

# Case Problem 3: Solution

(a) Work done by friction on the hammer moves from point 1 to point 2 is

$$W_f = -fs_{12}$$

where  $f$  is the friction force, and  $s_{12}$  is the distance between point 1 and point 2.

From conservation of energy,

$$K_1 + U_1 + W_{other} = K_2 + U_2$$

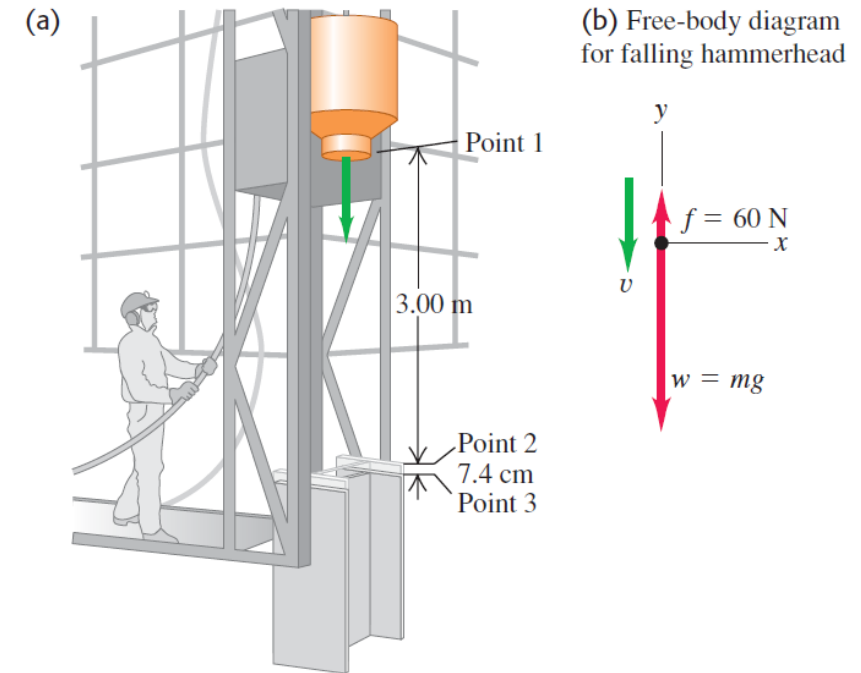
$$0 + mgs_{12} + W_f = \frac{1}{2}mv_2^2 + 0$$

but the hammer starts from rest, so  $K_1 = 0$ . Hence,

$$(mg - f)s_{12} = \frac{1}{2}mv_2^2$$

$$v_f = \sqrt{\frac{2(mg - f)s_{12}}{m}} = 7.55 \text{ ms}^{-1}$$

H.D. Young and R. A. Freedman, Sears and Zemansky's  
University Physics. 13E, Addison-Wesley, Boston, 2012.



## Case Problem 3: Solution

(b) The hammer comes to a full stop at point 3. Hence,  $K_3 = 0$ . From conservation of energy,

$$\begin{aligned} K_2 + U_2 + W_{other} &= K_3 + U_3 \\ K_2 + mgs_{23} + W_N + W_f &= 0 + 0 \\ -Nf_{23} - fs_{23} &= -K_2 - mgs_{23} \end{aligned}$$

where the magnitude of the normal force  $N$  is equal to that of the force exerted on the I-Beam by the hammer.

$$N = \frac{K_2 + mgs_{23} - fs_{23}}{s_{23}}$$

$$N = mg - f + \frac{2(mg - f)s_{12}}{s_{23}} = 79,000 \text{ N}$$

# Off-class practice question(s)



## Case Problem 4

You throw a 0.145-kg baseball straight up. Suppose your hand moves upward by 0.50 m while you are throwing the ball. The ball leaves your hand with an upward velocity of 20.0 m/s (a) Find the magnitude of the force (assumed constant) that your hand exerts on the ball. (b) Find the speed of the ball at a point 15.0 m above the point where it leaves your hand. Ignore air resistance.

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## Case Problem 4: Solution

- (a) Work done by the force exerted on the ball by your hand when moving the ball through a distance of  $s_{12} = 0.50 \text{ m}$  is  $W_F = F s_{12}$ .

Set  $U_1 = 0$ , and  $U_2 = m g s_{12}$ .

By conservation of energy,

$$\begin{aligned} K_1 + U_1 + W_F &= K_2 + U_2 \\ 0 + 0 + W_F &= \frac{1}{2} m v_2^2 + m g s_{12} \end{aligned}$$

Hence,  $(F - m g) s_{12} = \frac{1}{2} m v_2^2$

$$F = m \left( \frac{v_2^2}{2 s_{12}} + g \right) = 59.4 \text{ N}$$

## Case Problem 4: Solution

(b) Set  $U_2 = 0$ , and  $U_3 = mgh$  where  $h = 15 \text{ m}$ .

By conservation of energy,

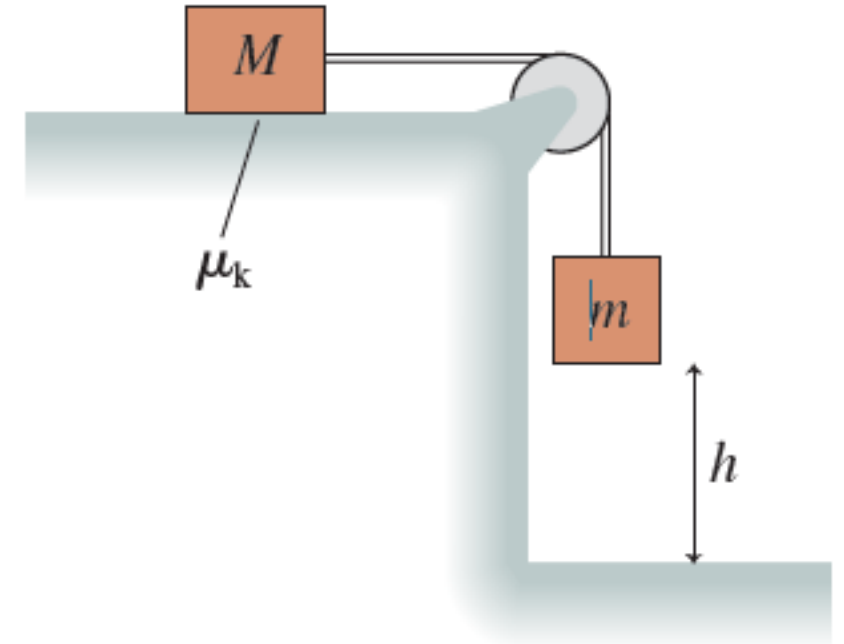
$$\begin{aligned} K_2 + U_2 + W_{other} &= K_3 + U_3 \\ \frac{1}{2}mv_2^2 + 0 + 0 &= \frac{1}{2}mv_3^2 + mgh \end{aligned}$$

$$v_3 = \sqrt{2 \left( \frac{v_2^2}{2} - gh \right)} = \sqrt{2 \left( \frac{20^2}{2} - 9.8 \times 15 \right)} \text{ ms}^{-1} = 10.3 \text{ ms}^{-1}$$

## Case Problem 5:

Use general law of conservation of energy to find an expression for the speed of the blocks (initially at rest) in figure just before it hits the floor if

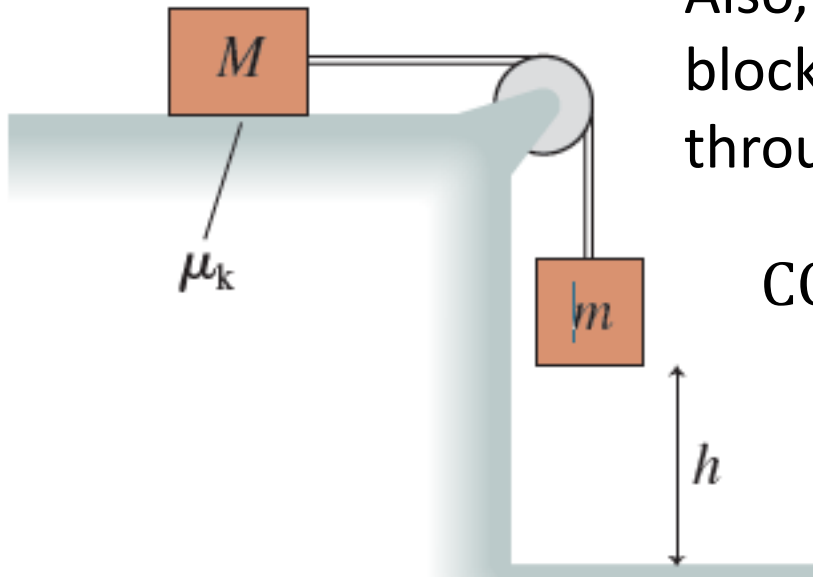
- a) the coefficient of kinetic friction for the block on the table is  $\mu_k$ .
- b) the table is frictionless.



R.D. Knight, Physics for Scientists and Engineers: A Strategic Approach. 4E, Pearson, Boston, 2016.

# Case Problem 5: solution

Apply the general law of conservation of energy,  $\sum(K_i + U_i) + W_{\text{other}} = \sum(K_f + U_f)$  to the two blocks as a system. From the question  $\sum K_i = 0$  (initially at rest), the only non-conservative force is the friction between block  $M$  with the table, so we have  $W_{\text{other}} = -|\vec{F}_k|h = -\mu_k M g \Delta x$ , where  $\Delta x = h$  if we assume the string connecting both blocks is not extendable.



Also, we are free to choose P.E. as zero at the ground level. P.E. of block  $M$  will not change since the height remains constant throughout. We have  $\sum(U_f - U_i) = -mgh$ .

$$\text{COE: } -\mu_k M g h = \frac{1}{2} (M + m) v_f^2 - m g h$$

$$\Rightarrow \frac{1}{2} (M + m) v_f^2 = m g h - \mu_k M g h$$

$$\Rightarrow v_f = \sqrt{\frac{2 g h (m - \mu_k M)}{(M + m)}} \quad \xrightarrow{\text{If frictionless}} \quad v_f = \sqrt{\frac{2 m g h}{(M + m)}}$$

R.D. Knight, Physics for Scientists and Engineers: A Strategic Approach. 4E, Pearson, Boston, 2016.